

Major Project Report

Learning Assistant Using LLMs and Knowledge Graphs

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GROUP ID - 81

Introduction



The Problem: Conventional Retrieval-Augmented Generation (RAG) systems lack the transparency, structural clarity, and traceability required for robust educational applications.



The Solution: This project introduces a Learning Assistant that creates personalized, traceable, and navigable knowledge paths for users.



Core Technology: The system moves beyond simple Q&A by integrating a custom-engineered data infrastructure, expandable and efficient database, a dual-LLM architecture, and dynamic Knowledge Graphs (KGs).



Key Motivations:

1. Solve lack of transparency and traceability in RAG.
2. Overcome static knowledge representation using dynamic KGs.
3. Address performance and scalability constraints of existing tools.

Literature Survey



RAG Systems: Our work extends current models by engineering custom chunk tracing and logging, ensuring all generated content is verifiable and traceable to its source.



Knowledge Representation: We advance the field by using small, local LLMs (via Llama.cpp) to *dynamically* generate Knowledge Graphs from retrieved context, moving beyond static, predefined structures, where every entity and relation is LLM-derived



Vector Indexing: We leverage state-of-the-art FAISS + HNSW for high-performance search, but innovate with a custom infrastructure that allows for incremental data management, avoiding costly full re-indexing.



Voice Interfaces: The project aligns with the modern EdTech trend towards natural and accessible interaction, integrating a full voice interface for conversational AI assistance.

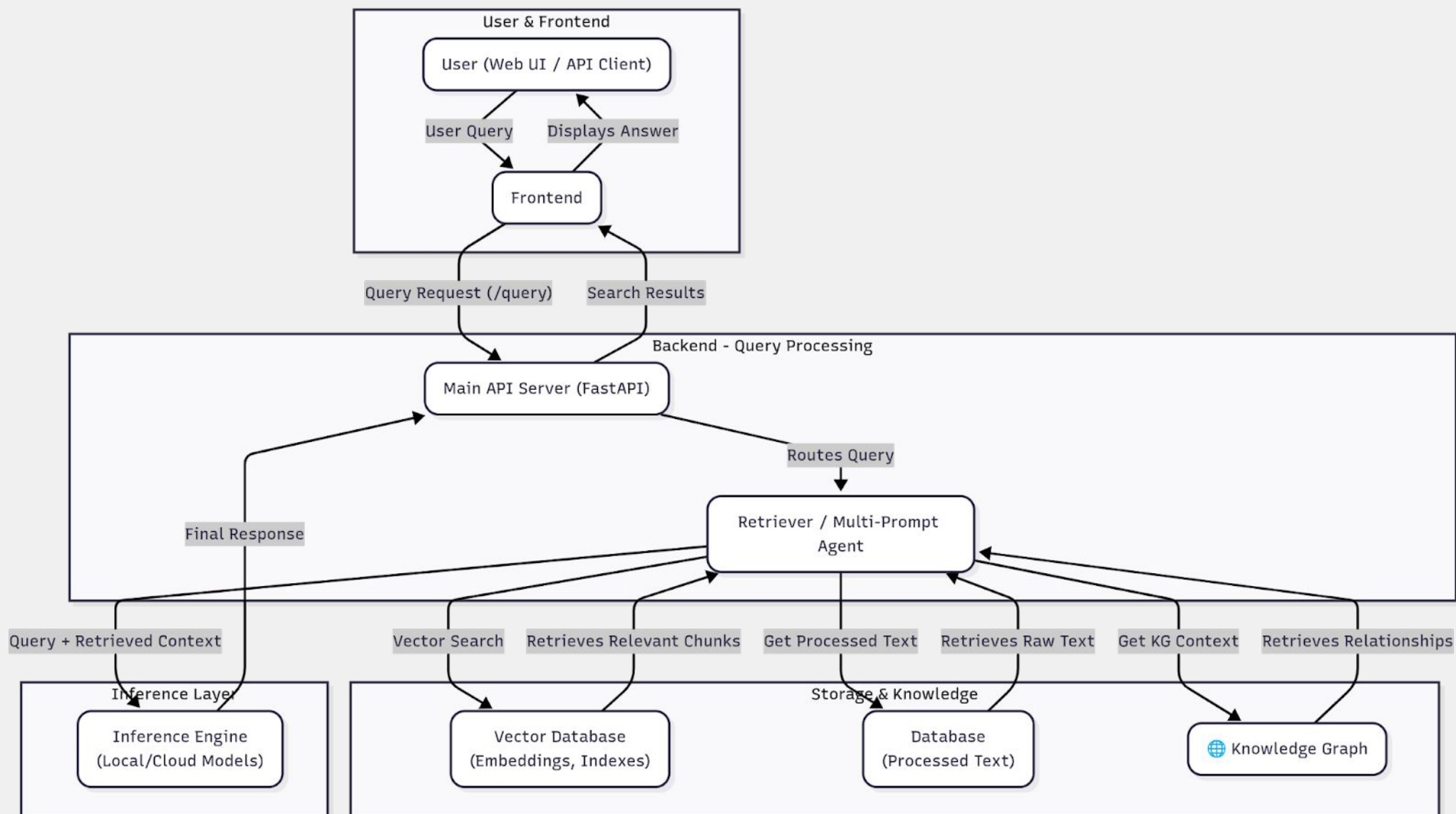
Objective & Methodology

Primary Objective: To design, implement, and validate a highly customizable, scalable Learning Assistant.

Core Technology: The system is built on a custom RAG framework, a dual-LLM architecture (Local + API), and a dynamic Knowledge Graph pipeline.

End Goal: To provide a fully traceable, efficient, and interactive knowledge system that serves as a robust, next-generation educational tool.

The diagram shows the full system flow, from user interaction (Frontend) to the Backend Orchestrator, Storage Layer, and Inference Layer.



Work Done: Harshit (2230026)

-  **Backend API & Server:** Led the development of the core FastAPI server, defining the API logic for all system endpoints.
-  **Data Ingestion Pipeline:** Built the custom pipeline to process and manage all incoming unstructured data.
-  **Data Warehouse:** Engineered the custom data warehouse to track content, manage incremental updates, and control the document lifecycle with soft deletion.
-  **Query & LLM Integration:** Implemented the "Deep Query" decomposition logic and integrated the high-speed Groq API for complex Q&A.
-  **Frontend Support:** Provided development support for integrating the frontend components with the backend API.

Work Done: Manodeep Ray (2230028)

-  **Vector DB & Retrieval:** Implemented the high-performance hybrid vector store, combining FAISS for rapid retrieval and HNSW for superior ANN search.
-  **Scalable Indexing:** Designed the vector store architecture for incremental processing, avoiding costly full re-indexing of the database.
-  **Chunking & Traceability:** Implemented the robust text chunking strategy with mandatory chunk tracing, linking embeddings to the exact source text.
-  **Dynamic Knowledge Graph:** Developed the core pipeline to dynamically construct Knowledge Graphs from retrieved context.
-  **Local Model Integration:** Integrated the Llama-CPP model to power the local processing for the Knowledge Graph pipeline.
-  **Agent Logic:** Developed of the logic for all Agentic system and generative endpoints.

Work Done: Nazheef Biswas (2230029)

-  **Interactive UI/UX Design:** Designed and developed the complete user interface, ensuring it is intuitive, responsive, and works across all devices.
-  **Chat Interface:** Built the user-facing chat component for seamless, natural language interaction with the AI.
-  **Knowledge Graph Visualization:** Integrated visualization tools to represent the generated KGs, allowing users to explore conceptual relationships.
-  **Generative Tools & Features:** Implemented the UI for all generative tools (Summaries, Quizzes, FAQs) and user-facing features.

Sources

+ Add

Discover

+ New Folder

Pre-uploaded Files

(12)

Harshit COA.pdf

Processed

emergent misalignment.pdf

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A_Brief_Introduction_To_AI.pdf

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Generate Outline

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Copy to Clipboard

Mexican-Food-1.pdf

Copy

Here is a hierarchical outline of the main topics and subtopics from the text:

I. Introduction to Tortillas

A. Flour tortillas

1. Now common-place

2. Displaced most corn tortillas

B. Versatility of tortillas

1. Used as wraps

2. Needed in various sizes and thicknesses

II. Old-line Food Manufacturers

A. La Preferida

1. History (founded in 1898)

2. Products (over 200 authentic Mexican food items)

3. Specialties (salsas, sauces, mole)

B. Goya Foods

1. History (founded in 1936)

2. Size and reach (largest Hispanic-owned food company in the US)

Studio

Tools

Notes 1

Voice Interface - Interact in:

हिन्दी বাংলা ગુજરાતી ಕನ್ನಡ മലയാളം मराठी ਪੰਜਾਬੀ தமிழ் తెలుగు

Coming soon

Outline

Summarize

FAQ

Quiz

Flashcards

Query

Studio output will be saved here. After adding sources, click to add Voice Interface, Study Guide, Mind Map, and more!

Results & Discussion

RAG Efficiency

The **Deep Query** method yielded a **20% increase in Recall** for complex, multi-faceted questions compared to the Standard Query. However, this improved coverage came at the cost of **2.5x the total computation time**.

LLM Latency & Quality

Llama-CPP (Local) was good for Entity Extraction, validating its use for the KG pipeline.

The **Groq API** was significantly faster for Summary Generation, proving essential for real-time generative tools. Summaries achieved >90% precision, while MCQs (~80%) require further prompt refinement.

Work to be carried out (in 8th Semester)



Robust Local Inference: Build a high-throughput inference engine with batching, streaming, and caching, running fully local LLMs on a dedicated GPU system.



Scalable Multi-User Deployment: Containerize all components using Docker and deploy on Kubernetes to manage horizontal scaling, ensuring the system can host multiple users with high availability.



Advanced Personalization:

1. Develop a UI for users to customize the Knowledge Graph generation.
2. Create an adaptive learning module (e.g., auto-generate flashcards for low quiz accuracy).



Full Voice Implementation: Integrate STT/TTS services and finalize WebSockets for real-time, conversational audio streaming.



Debate Function: Develop a new module where the AI can take opposing viewpoints on a topic to stimulate critical thinking.



Lecture & Transcription: Implement a feature for recording, transcribing, and automatically indexing lecture content into the knowledge base.

References (IEEE Format)

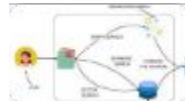
Ref	Publication
[1]	L. M. Lewis, P. Pérez, E. et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in <i>Proc. 34th Conf. on Neural Information Processing Systems (NeurIPS)</i> , 2020.
[2]	Y. A. Malkov and D. A. Yashunin, "Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs," <i>IEEE Trans. on Pattern Analysis and Machine Intelligence</i> , vol. 42, no. 4, pp. 824-836, 2020.
[3]	J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," <i>IEEE Trans. on Big Data</i> , vol. 7, no. 3, pp. 555-566, 2021.
[4]	Z. Pan, et al., "Unifying Large Language Models and Knowledge Graphs: A Survey," <i>arXiv preprint arXiv:2402.10237</i> , 2024.

Image Sources



<https://www.presse-service.de/data.aspx/medien/307787P.jpg>

Source: www.presse-service.de



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Source: levelup.gitconnected.com